NOUR LEMJID

Automotive Software Engineering Student

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PROFILE

Engineering student focused on embedded systems and electronics for automotive and aerospace applications. Interested in combining hardware, software and safety through projects involving driver-assistance systems (ADAS), battery management, emergency-signal detection and flight simulation. Aspiring to develop reliable technologies that improve road safety and future mobility.

EDUCATION

Engineering Cycle – Automotive Software | *2025–2028*

National Engineering School of Sousse (ENISo)

Baccalaureate

Secondary Education

PROJECTS

Flight Simulator

Academic project

Design of a flight simulator modelling aircraft dynamics (pitch, roll and yaw) with a real-time control interface.

- Tools: MATLAB/Simulink, Python, C++, Arduino and joystick.
- Expected outcome: a functional simulator responding to pilot commands in real time.

Driver Drowsiness Detection System

Academic project

Camera-based monitoring system intended to detect fatigue indicators such as prolonged eye closure, yawning and head inclination, then trigger visual and audible alerts.

- Tools: not yet specified.
- Expected outcome: real-time detection and automatic alerts.

Battery Management System (BMS)

Academic project

Design of a battery management system for lithium-ion cells, monitoring voltage, current and temperature, with protection, cell balancing and state-of-charge estimation.

- Tools: STM32/Arduino, voltage/current/temperature sensors, Proteus/KiCad and MATLAB/Simulink.
- Expected outcome: a working BMS with protection functions and state-of-charge display.

Emergency Signal Detection

Academic project

System concept for recognizing emergency signals such as sirens and flashing lights of priority vehicles to alert the driver.

- Tools: Python, FFT signal processing, OpenCV, machine learning, microphone and Raspberry Pi.
- Expected outcome: reliable signal detection and driver alerts.

Automatic Emergency Braking (AEB)

Academic project

Development concept for detecting obstacles ahead and assessing collision risk, with a warning and automatic braking response.

- Tools: ultrasonic sensor/LiDAR/radar, Arduino/STM32, MATLAB/Simulink, motors and driver for a prototype.
- Expected outcome: braking triggered according to time-to-collision (TTC).

EXPERIENCE & ACTIVITIES

Member | *Present*

Junior Entreprise ENISo (JEENISo)

- Participate in team challenges and contribute to Business Unit 4 (BU4) activities.

Member | *Present*
ODDCE Club ENISo

Member | *Present*
Robotics Club ENISo

- Participate in practical robotics activities using Arduino and explore embedded systems through teamwork.

Member
Tunisian Scouts

Member
Aerotronix

Member
AIESEC Tunisia

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB

Embedded systems: Arduino, STM32, ESP32

Tools & domains: MATLAB/Simulink, Proteus/KiCad, OpenCV, signal processing, embedded AI (self-study)

PERSONAL SKILLS

Analytical thinking, rigor, problem-solving, autonomy, teamwork, communication, time management and project organization.

LANGUAGES

Arabic: Native | French: B2 | English: B2

INTERESTS & CAREER GOALS

Electric and autonomous vehicles, aerospace and flight simulation, embedded artificial intelligence, power electronics and road safety. Short term: complete academic projects and obtain an internship in embedded systems or automotive engineering. Long term: become an ADAS/embedded systems engineer and contribute to safer, smarter and more electric vehicles.